

DS	Session	Searching	Question Information	Level 1 Challenge 8
----	---------	-----------	----------------------	-----------------------

Problem Description:

Ram's first job is a out of university is for a student's book publisher. They are currently publishing a line of "intelligent libs" books, and they need you to check their print-copy. Write a program that will take a sentence with placeholders in it, and substitute in for those placeholders words from the lists of example words the publisher has provided.

Your goal will be to substitute the appropriate words into the sentence from the list of provided words TWICE (keeping track of the words you have used) where:

- [N] = nouns
- [AV] = adverbs
- [V] = verbs
- [A] = adjectives

Input Format:

Line one will always be the line with the story. The order of the Nouns, Adverbs, Verbs and Adjectives will always be the same. Input stops when it reaches the word END alone on a line. You are guaranteed to always have enough words to substitute into the sentence TWICE (not repeating any words). The number of words given under each section will be random.

Output Format:

Print the output in a separate lines contains, Always take the TOP-most UNUSED word from a group when substituting. Output the final sentences when all substitutions are done.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

There once was a [AJ] [N] from [N] who [AV] [V] all day.
NOUNS
squirrel
London
tree
lettuce
ADVERBS
silently
quickly

Test Case 2

INPUT (STDIN)

There once was a [AJ] [N] from [N] who [AV] [V] all day.
NOUNS
singer
chennai
hotel
chettinad
ADVERBS
silently
quickly

happily
VERBS
skipped
ran
climbs
ADJECTIVES
delightful
homely
quain

EXPECTED OUTPUT

There once was a delightful squirrel from London who
silently skipped all day.
There once was a homely tree from lettuce who quickly ran
all day.

happily
sadly
VERBS
skipped
reads
ran
sings
climbs
ADJECTIVES
delightful
homely
surprisingly

EXPECTED OUTPUT

There once was a delightful singer from chennai who silently
skipped all day.
There once was a homely hotel from chettinad who quickly
reads all day.

▼ Mandatory Test Cases

Test Case 1

KEYWORD

int cl[CMDS]

Test Case 2

KEYWORD

char *lists[CMDS][MAXWORDS];

Test Case 3

KEYWORD

char *tokens[TOKENS]={"[N]","[AV]","[V]","[AJ]"};

Test Case 4

KEYWORD

char *cmds[CMDS]={
"NOUNS","ADVERBS","VERBS","ADJECTIVE
S","END"};

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

488

Test Case 3

NLOC

50

Searching8.c +

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <stdlib.h>
4
5 #define CMDS 5
6 #define MAXWORDS 100
7
8 void substitute(char *sentence, char *lists[CMDS][MAXWORDS], int *list_counts) {
9     int used[CMDS] = {0};
10
11     for (int pass = 0; pass < 2; pass++) {
12         char result[4000] = "";
13         char *ptr = sentence;
14
15         while (*ptr) {
16             if (*ptr == '[') {
17                 ptr++;
18                 char placeholder[10];
19                 int j = 0;
20                 while (*ptr && *ptr != ']') {
21                     placeholder[j++] = *ptr++;
22                 }
23                 placeholder[j] = '\\0';
24                 if (*ptr == ']') ptr++;
25
```

```

26     int cat = -1;
27     if (strcmp(placeholder, "N") == 0) cat = 0;
28     else if (strcmp(placeholder, "AV") == 0) cat = 1;
29     else if (strcmp(placeholder, "V") == 0) cat = 2;
30     else if (strcmp(placeholder, "AJ") == 0) cat = 3;
31
32     if (cat >= 0 && used[cat] < list_counts[cat]) {
33         strcat(result, lists[cat][used[cat]]);
34         used[cat]++;
35     } else {
36         strncat(result, "[", 1);
37         strncat(result, placeholder, strlen(placeholder));
38         strncat(result, "]", 1);
39     }
40 } else {
41     strncat(result, ptr, 1);
42     ptr++;
43 }
44 }
45

```

```
46     printf("%s\n", result);
47 }
48}
49
50int main() {
51     char line[4000];
52     char sentence[4000];
53     char *lists[CMDS][MAXWORDS];
54     int list_counts[CMDS];
55     char *cmds[CMDS] = {"NOUNS", "ADVERBS", "VERBS", "ADJECTIVES", "END"};
56
57     sentence[0] = '\0';
58
59     for (int i = 0; i < CMDS; i++) {
60         list_counts[i] = 0;
61         for (int j = 0; j < MAXWORDS; j++) {
62             lists[i][j] = NULL;
```

main | CH.SC.U4CSE25225 | Searching8.c

Ln 124, Col 1 Spaces: 4 UTF-8 CRLF C

```
64     }
65
```

```

66  int current_section = -1;
67
68  while (fgets(line, sizeof(line), stdin)) {
69      line[strcspn(line, "\n")] = 0;
70
71      if (strcmp(line, "END") == 0) {
72          if (sentence[0] != '\0') {
73              substitute(sentence, lists, list_counts);
74          }
75          break;
76      }
77
78      if (strchr(line, '[') != NULL) {
79          if (sentence[0] != '\0') {
80              substitute(sentence, lists, list_counts);
81
82              for (int i = 0; i < CMDS; i++) {
83                  for (int j = 0; j < list_counts[i]; j++) {
84                      free(lists[i][j]);
85                  }

```

```

86         }
87
88         for (int i = 0; i < CMDS; i++) {
89             list_counts[i] = 0;
90             for (int j = 0; j < MAXWORDS; j++) {
91                 lists[i][j] = NULL;
92             }
93         }
94     }
95
96     strcpy(sentence, line);
97     current_section = -1;
98     continue;
99 }
100
101 int is_header = -1;
102 for (int i = 0; i < CMDS - 1; i++) {
103     if (strcmp(line, cmds[i]) == 0) {
104         is_header = i;
105         break;

```

```

106     }
107 }
108
109 if (is_header >= 0) {
110     current_section = is_header;
111 } else if (current_section >= 0 && list_counts[current_section] < MAXWORDS) {
112     lists[current_section][list_counts[current_section]] = strdup(line);
113     list_counts[current_section]++;
114 }
115 }
116
117 for (int i = 0; i < CMDS; i++) {
118     for (int j = 0; j < list_counts[i]; j++) {
119         free(lists[i][j]);
120     }
121 }
122
123 return 0;
124 }

```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

There once was a [AJ] [N] from [N] who [AV] [V] all day.

NOUNS

squirrel

London

tree

lettuce

ADVERBS

silently

quickly

VERBS

skipped

ran

climbs

ADJECTIVES

delightful

homely

quain

END

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching8.exe

There once was a delightful squirrel from London who silently skipped all day.

There once was a homely tree from lettuce who quickly ran all day.

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

There once was a [AJ] [N] from [N] who [AV] [V] all day.

NOUNS

singer

chennai

hotel

chettinad

ADVERBS

silently

quickly

sadly

VERBS

skipped

reads

ran

sings

```
climbs
ADJECTIVES
delightful
homely
surprisingly
END
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching8.exe
There once was a delightful singer from chennai who silently skipped all day.
There once was a homely hotel from chettinad who quickly reads all day.
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```